

Tensor computations

# Comon's exact-rank conjecture for Hankel tensors

**Difficulty:** challenging **Importance:** interesting to specialist**Status:** Partially resolved

**Rating rationale:** Challenging because exceptional Hankel tensors require a uniform comparison of structured and unrestricted exact decompositions; specialist importance reflects this particular structured tensor class.

## Statement

For every  $m \geq 3$ ,  $n \geq 2$ , and  $h \in \mathbb{C}^{m(n-1)+1}$ , form the symmetric tensor

$$H_{i_1 \dots i_m} = h_{i_1 + \dots + i_m - m}, \quad 1 \leq i_j \leq n.$$

Is  $R(H) = R_{\text{sym}}(H)$  always true? Here  $R(H)$  is the least integer  $r \geq 0$  admitting

$$H = \sum_{j=1}^r u_{j,1} \otimes \dots \otimes u_{j,m}, \quad u_{j,k} \in \mathbb{C}^n,$$

whereas  $R_{\text{sym}}(H)$  is the least  $r$  admitting  $H = \sum_{j=1}^r c_j v_j^{\otimes m}$ , with  $c_j \in \mathbb{C}$  and  $v_j \in \mathbb{C}^n$ . Empty sums represent zero. No genericity or Vandermonde restriction is imposed.

## Relevance

This would justify symmetry-preserving decomposition lengths for every Hankel tensor, including exceptional data. Such structure appears in harmonic retrieval and structured tensor factorization.

## References

1. J. Nie and K. Ye, *Hankel Tensor Decompositions and Ranks*, SIAM J. Matrix Anal. Appl. 40 (2019). [DOI](#); [primary preprint](#), §6, Conjecture 6.3, p.16.
2. L. Qi, *Hankel Tensors: Associated Hankel Matrices and Vandermonde Decomposition*, 2014. [Primary preprint](#), §1, equation (1), and §4, Theorem 3, for Hankel structure and decomposition.

## Status check — 2026-09-10

Rechecked [Nie–Ye, Conjecture 6.3, Corollary 4.3 and Theorem 5.4](#), and searched for later Hankel Comon proofs and counterexamples. Generic even-order and order-three Hankel tensors are proved cases of the displayed universal equality; exceptional tensors and the remaining orders are not settled by those results. The paper already accounts for Shitov's unrestricted counterexample. No later full resolution was located.